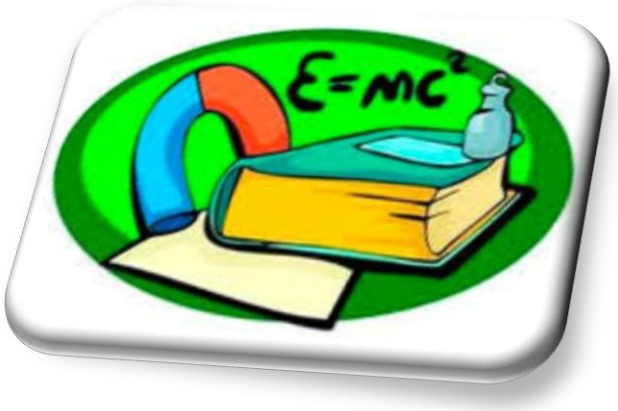
Fizika

9-sinf

Mavzu: **Masalalar yechish.**



19-mashq 3-masala

72 km/h tezlikda harakatlanayotgan avtomobilning 2 km yo‘ldagi benzin sarfini hisoblang. Avtomobilning quvvati 23 kW, FIK 23% ga teng.

Benzinning solishtirma yonish issiqligi 46 MJ/kg ga teng.

Berilgan:

$$\vartheta = 72\,\frac{km}{h} = 20\,\frac{m}{s}$$

$$S = 2\ km = 2000\ m$$

$$N_f = 23\ kW = 23000\ W$$

$$\eta = 23\ \%$$

$$q = 46\,\frac{MJ}{kg} = 46 \cdot 10^6\,\frac{J}{kg}$$

Topish kerak: $m - ?$

Formula:

$$A_t = qm\ ;\quad A_f = FS\ ;\quad \eta = \frac{A_f}{A_t} \cdot 100\ \%$$

$$\eta = \frac{FS}{qm} \cdot 100\ \% \Rightarrow m = \frac{FS \cdot 100\ \%}{\eta q}$$

$$N_f = F\vartheta \Rightarrow F = \frac{N_f}{\vartheta}$$

$$m = \frac{N_f S \cdot 100\ \%}{\vartheta \eta q}$$

Yechish: $$m = \frac{23000\ W \cdot 2000\ m \cdot 100\ \%}{20\frac{m}{s} \cdot 23\ \% \cdot 46 \cdot 10^6 \frac{J}{kg}} \approx 0{,}22\ kg$$

Javob: $m \approx 0,22\ kg$

19-mashq 5-masala

Ideal issiqlik mashinasidagi gaz isitgichdan olgan issiqlikning 60 % ini sovitkichga beradi.

Agar isitkichning temperaturasi 227℃ bo‘lsa, sovitkichning temperaturasi qanday bo‘lgan?

Berilgan:

$\eta = 40\ \%$

$t_1 = 227°\text{C}$

Topish kerak: $t_2 - ?$

Formula:

$T = t + 273\ ;\qquad \eta = \frac{T_1 - T_2}{T_1} \cdot 100\ \%$

$\eta T_1 = T_1 \cdot 100\ \% - T_2 \cdot 100\ \%$

$T_2 \cdot 100\ \% = T_1 \cdot 100\% - \eta T_1$

$$T_2 = \frac{T_1(100\ \% - \eta)}{100\ \%}$$

Yechish: $\text{T} = 227°\text{C} + 273 = 500\ \text{K}$

$$T_2 = \frac{500\ K \cdot (100\ \% - 40\ \%)}{100\ \%} = 300\ K\ ;\ t_2 = 300\ K - 273 = 27°\text{C}$$

Javob: $t_2 = 27°\text{C}$

19-mashq 7-masala

Gorizontal yo‘lda motosiklning dvigateli 60 km/h tezlikda 3,5 kW quvvatga erishadi.

Agar dvigatelning FIK 25 % bo‘lsa, motoroller $3{,}6\ l$ benzin sarflab, qancha yo‘lni bosib o‘tadi?

Benzinning solishtirma yonish issiqligi 46 MJ/kg, zichligi $0{,}7\ \frac{g}{cm^3}$.

Berilgan:

$$\vartheta = 60\,\frac{km}{h} = 16{,}7\,\frac{m}{s}$$

$$N_f = 3{,}5\ kW = 3500\ W$$

$$\eta = 25\ \%$$

$$V = 3{,}6\ l = 0{,}0036\ m^3$$

$$q = 46\,\frac{MJ}{kg} = 46 \cdot 10^6\,\frac{J}{kg}$$

$$\rho = 0{,}7\,\frac{g}{cm^3} = 700\,\frac{kg}{m^3}$$

Topish kerak: $S - ?$

Formula:

$$A_t = qm\ ;\ A_f = FS;\ \eta = \frac{A_f}{A_t} \cdot 100\ \%$$

$$\eta = \frac{FS}{qm} \cdot 100\ \% \Rightarrow S = \frac{\eta qm}{F \cdot 100\ \%}$$

$$N_f = F\vartheta\ \Rightarrow\ F = \frac{N_f}{\vartheta};\quad m = \rho V$$

$$S = \frac{\eta q \rho V \vartheta}{N_f \cdot 100\ \%}$$

$$S = \frac{\eta q \rho V \vartheta}{N_f \cdot 100\ \%}$$

Yechish: $$S = \frac{25\ \% \cdot 46 \cdot 10^6 \frac{J}{kg} \cdot 700 \frac{kg}{m^3} \cdot 0{,}0036\ m^3 \cdot 16{,}7\ m/s}{3500\ W \cdot 100\ \%} =$$

$$= 1\ 382\ 76\ m = 138{,}276\ km$$

Javob: $\boldsymbol{S = 138{,}276\ km}$

Masala

Sovitkichning absolyut temperaturasi isitkichning absolyut temperaturasining to‘rtdan bir qismiga teng.

Ideal issiqlik mashinasining FIK toping.

Berilgan:

$$T_2 = \frac{T_1}{4}$$

Topish kerak: $\eta - ?$

Formula:

$$\eta = \frac{T_1 - T_2}{T_1} \cdot 100\ \%$$

Yechish: $\eta = \frac{T_1 - \frac{T_1}{4}}{T_1} \cdot 100\ \% = \frac{3}{4} \cdot 100\ \% = 75\ \%$

Javob: $\eta = 75\ \%$

Masala

FIK 40 % bo‘lgan ideal issiqlik mashinasi isitkichdan 10 kJ issiqlik oladi.

Sovitkichga berilgan issiqlik miqdori qanchaga teng (kJ)?

Berilgan:

$\eta = 40\ \%$

$Q_1 = 10\ kJ$

Topish kerak: $Q_2 - ?$

Formula:

$$\eta = \frac{Q_1 - Q_2}{Q_1} \cdot 100\ \%$$

$$\frac{\eta Q_1}{100\%} = Q_1 - Q_2$$

$$Q_2 = Q_1 - \frac{\eta Q_1}{100\%}$$

Yechish: $Q_2 = 10\ kJ - \frac{40\% \cdot 10\ kJ}{100\%} = 6\ kJ$

Javob: $\boldsymbol{Q_2 = 6\ kJ}$

Mustaqil bajarish uchun topshiriqlar

1. 19-mashq 4, 6, 8-masalalarni yechish. (90-91-bet)

2. 91-betdagi 7, 8, 10-testlarni yechish.